Reference Notes

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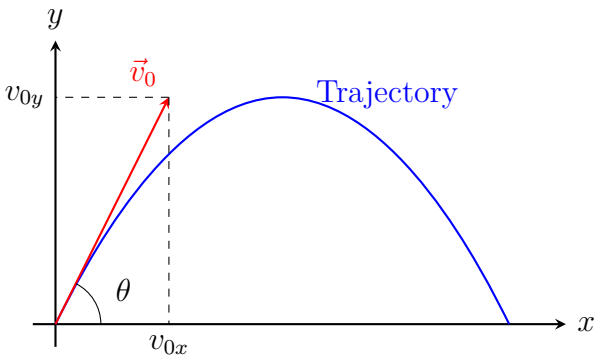
The derivation of the projectile motion trajectory equation demonstrates that an object in free fall follows a parabolic path. Below is the LaTeX code for the trajectory diagram using `tikzpicture`, followed by the mathematical derivation.

LaTeX Diagram Code (TikZ):



Derivation of the Trajectory Equation:

1. **Horizontal Motion (Constant Velocity):** There is no acceleration in the horizontal direction ($a_x = 0$). The horizontal position x at any time t is determined by the initial horizontal velocity component.

$$x = (v_0 \cos\theta)t$$

Solving for time t:

$$t = \frac{x}{v_0 \cos\theta}$$

2. **Vertical Motion (Constant Acceleration):** The object experiences constant downward acceleration due to gravity ($a_y = -g$). The vertical position y at time t is:

$$y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$$

3. **Substitution:** Substitute the expression for t from the horizontal equation into the vertical equation to eliminate time and relate y directly to x:

$$y = (v_0 \sin\theta)\left(\frac{x}{v_0 \cos\theta}\right) - \frac{1}{2}g\left(\frac{x}{v_0 \cos\theta}\right)^2$$

4. **Simplification:** Cancel v_0 in the first term and use the trigonometric identity $\frac{\sin\theta}{\cos\theta} = \tan\theta$:

$$y = x\tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}$$

This final equation takes the form $y = Ax - Bx^2$ (where A and B are constants), which is the standard mathematical equation for a parabola.